

Topic 10: Reversible and irreversible changes

Students should:		Guidance <i>iPrimary Curriculum reference</i>
10.1	explain, with examples, that some mixtures can be separated using sieving, filtration or magnetism	this builds on 9.6 and <i>Year 3: Magnets</i> examples may include combinations of sugar/salt/flour/sand/iron filings/rice grains/pebbles/stones <i>Year 6: Reversible and irreversible change</i>
10.2	understand the terms <i>dissolving</i> , <i>solution</i> , <i>solvent</i> and <i>solute</i>	<i>Year 6: Reversible and irreversible change</i>
10.3	explain how a solute can be recovered from a solution by evaporating the solvent	for example heating salt solution, this builds on 9.5 <i>Year 6: Reversible and irreversible change</i>
10.4	understand that <i>melting</i> , <i>freezing</i> , <i>evaporation</i> and <i>condensation</i> are changes of state	<i>Year 6: Reversible and irreversible change</i>
10.5	know that changes of state require changes of temperature	<i>Year 6: Reversible and irreversible change</i>
10.6	know the role of evaporation and condensation in the water cycle	limited to a simple water cycle comprising of evaporation, condensation, precipitation, sea, clouds, land <i>Year 6: Reversible and irreversible change</i>
10.7	understand that dissolving, mixing and changes of state are reversible changes	<i>Year 6: Reversible and irreversible change</i>
10.8	explain that some changes result in the formation of new materials, and that this kind of change is not usually reversible	<i>Year 6: Reversible and irreversible change</i>
10.9	know about simple irreversible changes, for example iron nails rusting, wax burning, wood burning, bread cooking	<i>Year 6: Reversible and irreversible change</i>
10.10	know that the changes observed when acid and bicarbonate of soda are mixed are evidence that new materials are formed	<i>Year 6: Reversible and irreversible change</i>